

DATA CLEANING CHANGE LOG

Project	Data Analytics – Project 1 Data Cleaning & Preparation	Batch	2026
Organization	DecodeLabs	Date	2026-06-11
Dataset	Dataset_for_Data_Analytics.xlsx (1,200 rows × 14 cols)	Tool	Python 3 – pandas + openpyxl

Executive Summary

This document serves as the official audit trail for all data transformations applied to the DecodeLabs Dataset during Project 1. The dataset contained **1,200 records** and **14 columns** representing e-commerce order transactions. Three cleaning phases were executed: **Strategic Imputation** of missing values, **Integrity Audit** for duplicates and unique identifiers, and **Format Standardization** for dates, numerics, and text fields. All 1,200 records were preserved. Zero records were deleted. The dataset is now production-ready.

1,200	0	8	100%
Records Preserved	Records Deleted	Changes Logged	Completion Rate

Verification Gate — Project 2 Threshold

Requirement	Target	Result	Gate
Duplicate OrderIDs	0% error rate	0 duplicates found	PASSED ✓
Duplicate TrackingNumbers	0% error rate	0 duplicates found	PASSED ✓
Incorrect Date Formats	0% error rate	All dates → YYYY-MM-DD	PASSED ✓
Missing Values	0 nulls	309 imputed (NO_COUPON)	PASSED ✓

Detailed Change Log

CR001		Phase 1 – Strategic Imputation		● Resolved
Column(s):	CouponCode	Records Affected:	309 records	
Issue Identified:	309 null values (25.8% of dataset)			
Action Taken:	Filled with literal string "NO_COUPON"			
Justification:	Listwise deletion would remove 25.8% of records, reducing statistical power. "NO_COUPON" is a valid, meaningful business state indicating the customer did not apply a coupon.			
CR002		Phase 2 – Integrity Audit		● Verified ✓
Column(s):	All Columns	Records Affected:	0 records removed	
Issue Identified:	Full-row duplicate check performed			
Action Taken:	drop_duplicates(keep="first") applied			
Justification:	Duplicate rows inflate transaction counts and distort revenue aggregations. Zero duplicates were found — dataset confirmed clean.			
CR003		Phase 2 – Integrity Audit		● Verified ✓
Column(s):	OrderID	Records Affected:	0 records	
Issue Identified:	Unique identifier integrity check			
Action Taken:	Verified: 0 duplicate OrderIDs across 1,200 records			
Justification:	OrderID is the primary key. Duplicate IDs would cause incorrect JOIN results and inflated order counts. Gate requirement: 0% error rate — PASSED.			
CR004		Phase 2 – Integrity Audit		● Verified ✓
Column(s):	TrackingNumber	Records Affected:	0 records	
Issue Identified:	Unique identifier integrity check			
Action Taken:	Verified: 0 duplicate TrackingNumbers across 1,200 records			
Justification:	Each shipment must have a unique tracking number. Duplicates would break logistics integrations.			

CR005		Phase 3 – Format Standardization		● Resolved
Column(s):	Date	Records Affected:	1,200 records	
Issue Identified:	Stored as datetime64 — no enforced string format			
Action Taken:	Converted all dates to ISO 8601 string format: YYYY-MM-DD			
Justification:	ISO 8601 is the international standard for date representation. Ensures compatibility across Excel, SQL, Python, and BI tools. Gate requirement: 0% formatting errors — PASSED.			
CR006		Phase 3 – Format Standardization		● Resolved
Column(s):	UnitPrice, TotalPrice	Records Affected:	29 records corrected	
Issue Identified:	29 rows with floating-point precision errors (e.g. 769.3799999...)			
Action Taken:	Applied .round(2) to UnitPrice and TotalPrice columns			
Justification:	Floating-point binary representation causes stored values to exceed 2 decimal places. Currency fields must be precise to 2 decimal places (cents) to avoid financial calculation errors.			
CR007		Phase 3 – Format Standardization		● Verified ✓
Column(s):	Product, OrderStatus, PaymentMethod, ReferralSource, ShippingAddress	Records Affected:	0 issues found — verified clean	
Issue Identified:	Potential leading/trailing whitespace in all text columns			
Action Taken:	Applied str.strip() to all categorical and text columns			
Justification:	Invisible whitespace causes GROUP BY mismatches in SQL and value_counts discrepancies in Python (e.g. "Shipped" vs " Shipped" treated as different categories). Pre-emptive fix.			
CR008		Phase 3 – Format Standardization		● Resolved
Column(s):	CouponCode	Records Affected:	1,200 records standardized	
Issue Identified:	Coupon codes require uniform casing for lookup consistency			
Action Taken:	Applied str.upper() + str.strip(); null values already imputed as "NO_COUPON"			
Justification:	Coupon code lookups are case-sensitive in most e-commerce systems. Mixed case (e.g. "save10" vs "SAVE10") would cause failed redemption checks and incorrect discount analysis.			

Cleaning Summary — Before vs After

Metric	Before Cleaning	After Cleaning	Change Ref
Total Records	1,200	1,200 (0 deleted)	—
Missing CouponCode	309 null values (25.8%)	0 nulls (imputed NO_COUPON)	CR001
Duplicate Rows	0	0 (verified)	CR002
Duplicate OrderIDs	0	0 (verified)	CR003
Duplicate TrackingNumbers	0	0 (verified)	CR004
Date Format	datetime64 object	YYYY-MM-DD (ISO 8601)	CR005
TotalPrice Precision	29 rows with float errors	All values .round(2)	CR006
String Whitespace	Unchecked	Stripped (5 columns)	CR007
CouponCode Casing	Mixed / Null	UPPERCASE + NO_COUPON	CR008

This change log certifies that the cleaned dataset meets all Project 1 requirements and passes the Verification Gate for Project 2 access. All transformations are documented, justified, and reproducible.